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**To:** The Editor-in-Chief and the Editorial Board
*Forecasting* (ISSN 2571-9394), MDPI
Section: *Forecasting in Economics and Management*
Topical Collection: *Supply Chain Management Forecasting*

**Subject:** Submission of an original research article entitled *"Automated Demand Forecasting Through Feature-Based Model Selection: Development and Validation of a Computational Tool"*

Dear Editor,

On behalf of all co-authors, I am pleased to submit the above manuscript for consideration for publication as an *Article* in *Forecasting*, within the section *Forecasting in Economics and Management* and the Topical Collection *Supply Chain Management Forecasting*.

**Motivation.** Many manufacturing firms must forecast demand from short histories, typically 24–48 monthly observations, and in practice still rely on hand-computed moving averages with no systematic evaluation of forecast error. Automating method selection in this regime is not straightforward: structural tests for trend and seasonality are unreliable in short, autocorrelated samples, and accuracy claims are easily overstated when the data used to select a model are reused to report its performance. Our study addresses both problems directly, starting from a diagnosis of the forecasting process of an industrial reference company in the plastics manufacturing sector.

**Contribution.** We specify, implement, and validate a forecasting protocol and an open-source computational tool for short industrial demand histories, built on three safeguards: (i) a sequential structural classification procedure (seasonality tested before trend) calibrated for short, autocorrelated series; (ii) a nested tuning/evaluation split complemented by an outer block that keeps the choice of the *method itself* separate from the benchmark comparison used to judge it; and (iii) mandatory naive and seasonal-naive benchmarks, with the Mean Absolute Scaled Error (MASE) as the primary ranking criterion. The tool closes the loop between forecasting and decision-making by returning empirical prediction intervals together with the corresponding safety stock and reorder point. Rather than asserting the value of each safeguard, the manuscript measures how much the reported conclusions change when each one is omitted.

**Principal findings.**

- The calibrated sequential test reduces the simulated false-positive rate of trend detection from 74.4% to a mean of 8.9%, while a circular selection-versus-naive comparison is shown to inflate the apparent win rate from 63% to 100%.
- On an outer block held out from both tuning and method selection, the tool reduces per-series MASE relative to the incumbent three-period moving average by a median of 19.9% and outperforms it on 80% of 40 series drawn from a synthetic panel that reproduces the reference company's demand structure.
- On 150 public short series from the M3-Monthly panel, the tool outperforms the naive benchmark on 63% of series (Wilcoxon $W = 2127.0$, $p < 0.001$).
- Against Prophet and LightGBM, evaluated under the same leakage-free protocol on 50 series spanning five structural regimes, the tool attains the lowest overall median MASE (0.747 vs. 0.785 and 0.893). We report transparently that its advantage over Prophet, although significant ($p = 0.043$), is not robust to bootstrap resampling, and that a global LightGBM model performs best on flat and zero-inflated series.

**Fit with *Forecasting*.** The manuscript lies at the core of the journal's scope: it concerns forecast model selection and forecast evaluation methodology (scaled error measures, rolling-origin validation, mandatory benchmarks), building on the feature-based model selection literature (e.g., FFORMA and meta-learning approaches) and on the evidence of the M-competitions, and it translates that methodology into a forecasting support tool for practitioners. It is particularly well aligned with the Topical Collection *Supply Chain Management Forecasting*, which emphasizes practice-oriented studies of demand forecasting, intermittent demand, and forecast error and cost metrics: our work targets industrial demand with project-driven volatility and zero periods, adopts a zero-robust primary metric, and links forecast uncertainty explicitly to inventory-policy parameters.

**Reproducibility and transparency.** All code, an automated test suite of 276 tests, and eight single-command, fixed-seed scripts that regenerate every quantitative result are publicly available at `https://github.com/GIUSEPPESAN21/Forecasting`; a versioned release will be archived on Zenodo with a permanent DOI upon

acceptance. The reference company's sales records are proprietary; a synthetic panel reproducing their structural characteristics is provided so that all analyses can be executed end to end.

**Declarations.** We confirm that:

- this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere;
- all authors have read and approved the manuscript and agree with its submission to *Forecasting*;
- the manuscript has not been previously submitted to any MDPI journal;
- the authors declare no conflicts of interest, and the research received no external funding;
- the study involves no human participants, animals, or identifiable personal data, and raises no dual-use research concerns;
- the use of a generative AI system is disclosed in the Methods section and in the Acknowledgments in accordance with MDPI policy; the authors reviewed all outputs and take full responsibility for the content of the manuscript.

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Thank you for considering our manuscript. We look forward to the reviewers' comments.

Sincerely,

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